

# TIANRUI WU

Physics & Mathematics Student, Quantitative Researcher

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## Education

### University of Arizona

Double Major in Physics and Mathematics | GPA 3.87/4.0

Jan. 2026 – May 2027

Tucson, Arizona

### Duke University

Double Major in Physics and Mathematics, Minor in CS | GPA 3.92/4.0, Physics GPA 4.0

Aug. 2021 – Dec. 2024

Durham, North Carolina

## Research Experience

### BAO Reconstruction with Photometric Redshift Smearing

Jan. 2026 – Present

Research Assistant, Mentor: Prof. Tim Eifler

Arizona Cosmology Lab, University of Arizona

- Building a baryon acoustic oscillation (BAO) analysis pipeline that quantifies the impact of photometric redshift smearing on standard-ruler calibration, a key systematic for next-generation cosmological surveys including Roman and Rubin.
- Developing a forward-model framework that propagates photo- $z$  uncertainty through BAO reconstruction, with first-author publication in preparation.

### Predicting Power Spectrum Window Function with Neural Networks

Jun. 2024 – Aug. 2024

Summer Research Fellow (SURF), Mentor: Dr. Olivier Doré

SPHEREx Team, Caltech

- Built a ResUNet convolutional neural network in PyTorch to deconvolve instrument window-function effects from galaxy power spectra, achieving sub-percentage prediction error on validation data.
- Owned the full ML pipeline: lognormal mock catalog generation with real SPHEREx survey masks, distributed training on Linux HPC, and validation against analytic baselines.
- Results enhance inference accuracy for local primordial non-Gaussianity  $f_{\text{NL}}^{\text{loc}}$ , improving constraints on early-universe inflation dynamics.
- Delivered project proposal, two interim reports, final presentation, and final report to the SPHEREx faculty committee.

### Weak Lensing Cosmology with Survey Synergies

May 2022 – Dec. 2024

Research Assistant, Mentor: Prof. Michael A. Troxel

Cosmology Group, Duke University

- Engineered a Python pipeline for joint statistical analysis on 120 TB datasets from Roman Space Telescope and Rubin Observatory simulations, constructing two-point correlation functions and implementing Bayesian inference via MCMC chains to constrain cosmological parameters.
- Evaluated mode-based strategies and ground-space survey synergies for photometric redshift calibration; new methods yielded a 20% increase in Figure of Merit and tighter constraints on cosmic homogeneity.

### Calibration of Distance Ladders with Synthetic Stellar Populations

Sep. 2023 – Aug. 2024

Research Assistant, Mentor: Prof. Daniel Scolnic

Cosmology Group, Duke University

- Developed a Python pipeline to generate synthetic stellar catalogs and systematically extract features from Hertzsprung–Russell diagrams; integrated novel distance calibration methods into a unified framework for cross-team comparisons.
- Compared JWST distance measurement results across recent literature; quantified galaxy–dust correlations using latest SDSS data via SQL queries and Python analysis scripts. Contributed to ApJ publication (listed below).

### Quantifying Wetland Carbon Emissions in the Southeastern US

May 2023 – Aug. 2023

Team Lead, Mentor: Prof. Wenhong Li

Data+ Program, Duke University

- Led a four-person team: employed climatology-based adaptive missing-value imputation, trained and optimized a random forest model via grid search, mentored teammates on SVM and MLP models, and delivered a layered Shiny web application and final presentation.

## Publications

Wu, T., Paillas, E., et al., *BAO Reconstruction with Photometric Redshift Smearing*, in preparation (2026).

Riess, A.G., Scolnic, D., et al. (Wu, T. included), *JWST Validates HST Distance Measurements: Selection of Supernova Subsample Explains Differences in JWST Estimates of Local  $H_0$* , arXiv:2408.11770, ApJ (2024).

Alarcón, A., Aldoroty, L., Beltz-Mohrmann, G., et al. (Wu, T. included), *OpenUniverse2024: A Shared, Simulated View of the Sky for the Next Generation of Cosmological Surveys*, Monthly Notices of the Royal Astronomical Society, 544(4), 3799–3823 (2025).

## Talks & Presentations

SPHEREx Science Team Meeting (Face-to-Face E4), Cosmology Session, IPAC, Caltech — presented BAO measurements. Jun. 2026.

SO/NSF NOIRLab Joint Colloquium, Steward Observatory, University of Arizona — presented the BAO analysis segment of a SPHEREx team colloquium led by Prof. E. Krause and Prof. T. Eifler. Apr. 2026.

SURF Final Presentation, SPHEREx Team, Caltech — “Predicting Power Spectrum Window Function with Neural Networks.” Aug. 2024.

Industry Experience

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| <b>Quantitative Researcher</b>   <i>Qrigin Capital Co., Ltd. (\$15M prop shop), Remote</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Aug. 2025 – Present</b>   |
| <ul style="list-style-type: none"><li>• <b>Proprietary Strategy:</b> Developed a short-term forecasting engine using an LSTM-based neural network with phase-space features; managed the full MLOps lifecycle with MLFlow. Deployed a live HFT strategy delivering post-fee Sharpe &gt; 10 with consistent profit over &gt; 3 months on Coinbase.</li><li>• <b>Execution Research:</b> Architected execution algorithms (TWAP, VWAP, IS) using microstructure signals across major CEXs (OKX, Coinbase, Binance), achieving ~2 bps slippage reduction over 5 months.</li><li>• <b>Factor Research:</b> Achieved a 3.1% lift in live-model win rates by engineering 200+ multi-frequency factors (OHLCV, orderbook, derivatives) with automated decay analysis and cross-sectional ranking.</li></ul> |                              |
| <b>Quantitative Research Intern</b>   <i>InsightCheck Investment (\$300M AUM), Shenzhen</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Feb. 2025 – Aug. 2025</b> |
| <ul style="list-style-type: none"><li>• <b>Alpha Research:</b> Designed fundamental and microstructure-based factors for A-share cross-sectional portfolio optimization using Polars; validated 62 intraday factors with daily IC &gt; 0.01.</li><li>• <b>Orderbook Engine &amp; Infrastructure:</b> Proactively built an L3 orderbook reconstruction system in Python, then migrated it to C++. Developed a distributed backtesting engine with Dask, reducing evaluation cycles to 15 min (10× speedup).</li></ul>                                                                                                                                                                                                                                                                                 |                              |
| <b>Teaching Assistant, SAGE Facilitator</b>   <i>Math Department, Duke University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Aug. 2022 – Apr. 2024</b> |
| <ul style="list-style-type: none"><li>• Assisted 40+ students weekly with Multivariable Calculus, Linear Algebra, and Advanced Probability during two 3-hour Help Room shifts; led two 1-hour SAGE discussion sessions weekly for Multivariable Calculus. Rated “Excellent” (30/30) by supervisor and peers in annual TA reviews for 2023 and 2024.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                        |                              |

Skills & Coursework

**Tools:** PyTorch, MLFlow, NumPy, pandas, Polars, astropy, Matplotlib, Dask; Linux HPC.  
**Skills:** Deep learning, Bayesian inference, statistical modeling, market microstructure, optimal execution.  
**Languages:** Python (expert), Rust, C/C++, Java, SQL, MATLAB, JavaScript, L<sup>A</sup>T<sub>E</sub>X | English, Mandarin.  
**Selected Coursework:** Cosmology\*, Stellar Astrophysics\*, General Relativity\*, Abstract Algebra\*, Electrodynamics, Computational Physics, Quantum Theory, Statistical Inference, Probability, Differential Geometry, Machine Learning\*, Data Structures and Algorithms.

Honors & Awards

- Global Wildcat Scholarship, University of Arizona (2026)
- Faculty Scholars 2024 Finalist, Duke University — sole physics nominee (highest undergraduate honor)
- Jocelyn Bell Burnell Outstanding Leadership Scholarship, Society of Physics Students
- Inductee, Sigma Pi Sigma, National Physics Honor Society (2024)
- Dean’s List (Full-Time), University of Arizona (Spring 2026)
- Dean’s List with Distinction, Duke University (Fall 2021, Spring 2023, Fall 2023, Fall 2024)
- Physics & Astronomy 3rd Grand Award, 2021 Regeneron International Science and Engineering Fair (ISEF)
- Top Gold Medalist, British Physics Olympiad (2021)

Leadership & Outreach

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| <b>Stargazing Devils Astronomy Club</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Sep. 2021 – Dec. 2024</b> |
| <i>Founder, President</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Duke University</i>       |
| <ul style="list-style-type: none"><li>• Founded the first Astronomy Club at Duke. Organized weekly observatory open houses with 100+ attendees; hosted three comet events, four meteor showers, and three eclipse events. Served as NASA Partner Eclipse Ambassador.</li><li>• Secured funding and organized trips to the 2023 and 2024 solar eclipses for 14 students; organized a campus watch party with 300+ participants. Led 20+ astrophotography events using the Skynet platform.</li></ul> |                              |
| <b>Society of Physics Students</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Jun. 2022 – Dec. 2024</b> |
| <i>President (2024), Co-President (2023)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Duke University</i>       |
| <ul style="list-style-type: none"><li>• Revived the chapter post-pandemic: organized four research panels, six student–faculty seminars, three lab tours, and launched MATLAB/L<sup>A</sup>T<sub>E</sub>X workshops, grad school info panels, and a Women’s mentorship program.</li><li>• Organized the 2023 SPS Zone 5 meeting with 100+ participants; won the 2023 Outstanding Chapter Award.</li></ul>                                                                                           |                              |
| <b>American Physical Society</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Jun. 2022 – May 2024</b>  |
| <i>APS Student Ambassador</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <i>Remote</i>                |
| <ul style="list-style-type: none"><li>• Recruited APS members; organized physics study sessions, manuscript workshops, and publication parties. Hosted the Physics Undergraduates Learning and Sharing Experiences (PULSE) Webinar Series as Technology Chair.</li><li>• Invited to the 2023 and 2024 APS Leadership Meetings; advocated on physics community policy priorities at US Congressional Visits Day.</li></ul>                                                                           |                              |
| <b>Duke Vertices Science Journal</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Sep. 2021 – Aug. 2024</b> |
| <i>Senior Editor, Peer Reviewer</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>Duke University</i>       |
| <ul style="list-style-type: none"><li>• Reviewed 2 STEM manuscripts per month; implemented a synopsis system for general audiences yielding a 210% increase in website traffic; invited by the editor-in-chief to join the executive team.</li></ul>                                                                                                                                                                                                                                                |                              |

\* graduate-level course